Master’s thesis defense · DTU Electro

Design and Implementation of a High-Frequency Charge-Pump PFC Resonant Converter for a USB-C Power Supply

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A USB-C supply for a changing load

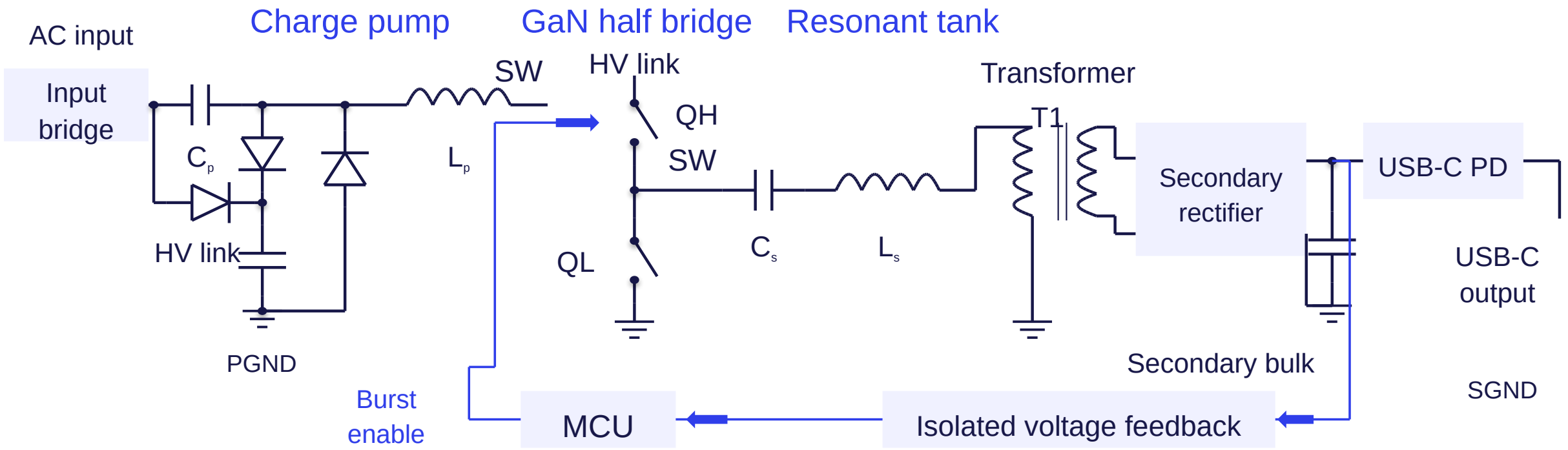
No load → **Device connected** → **Load changes** → **Device disconnected**

Continuous output goal: 100 W target · 65 W minimum

- Integrated the circuit and built prototypes
- Developed secondary-voltage control
- Measured electrical and thermal behavior

One half bridge drives both power paths

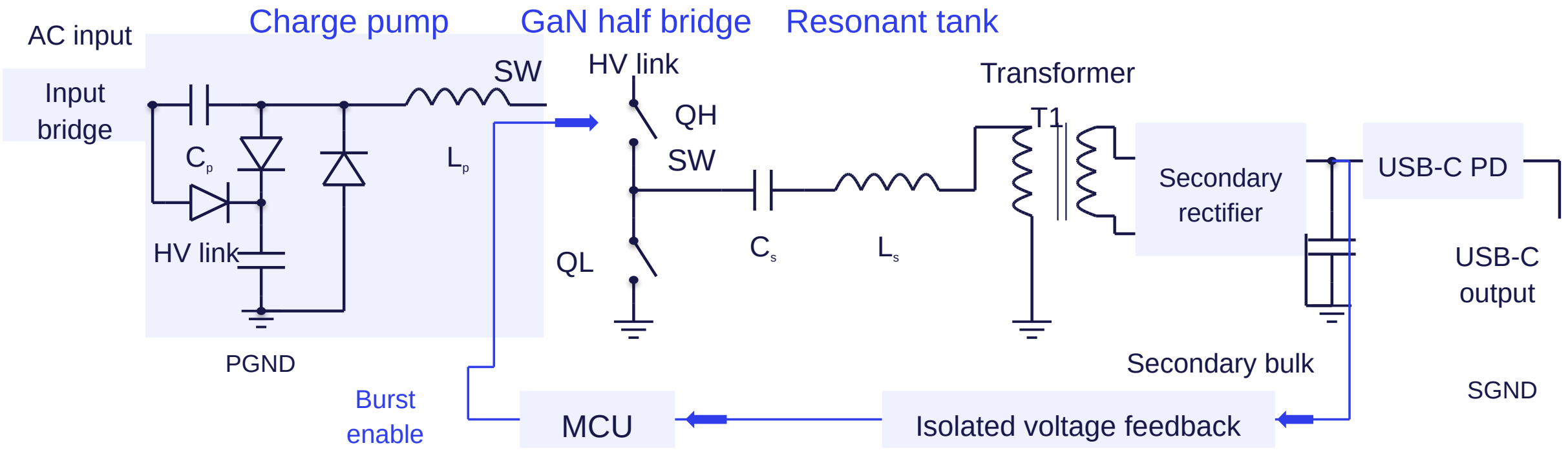
V2 power circuit



MCU regulates secondary bulk voltage

PD stage regulates selected USB-C output voltage

Charge transfer shapes the input current

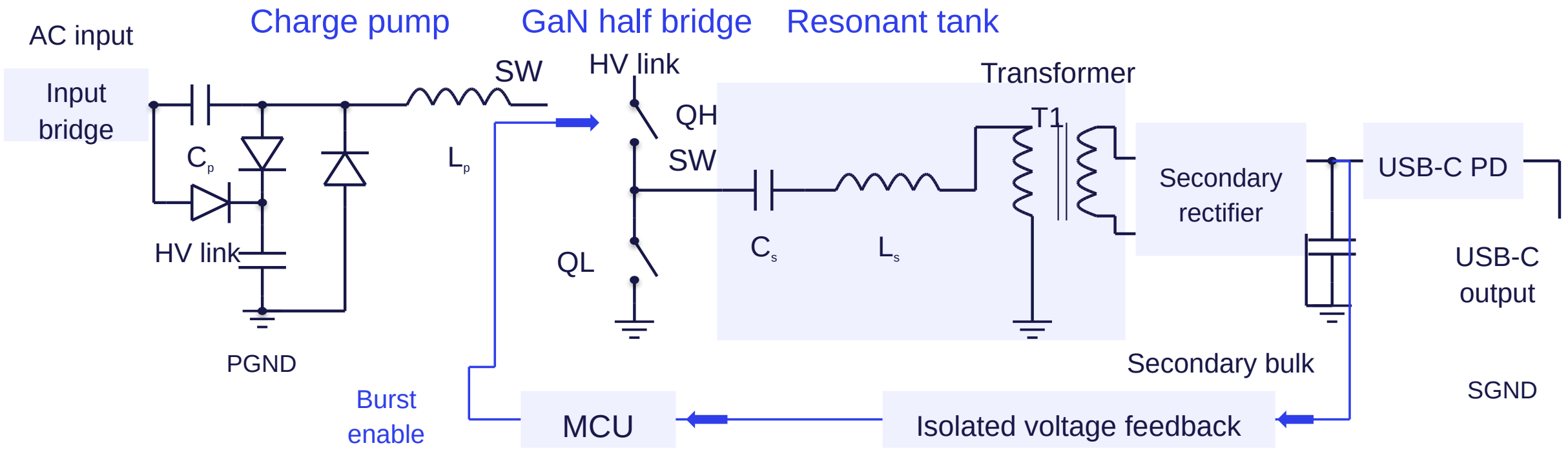


$$\bar{I}_{rec} \approx b\, f_s\, C_p\, V_{rec}$$

b: local enabled-cycle fraction

Burst timing changes the current envelope

Resonant transfer runs near 1 MHz



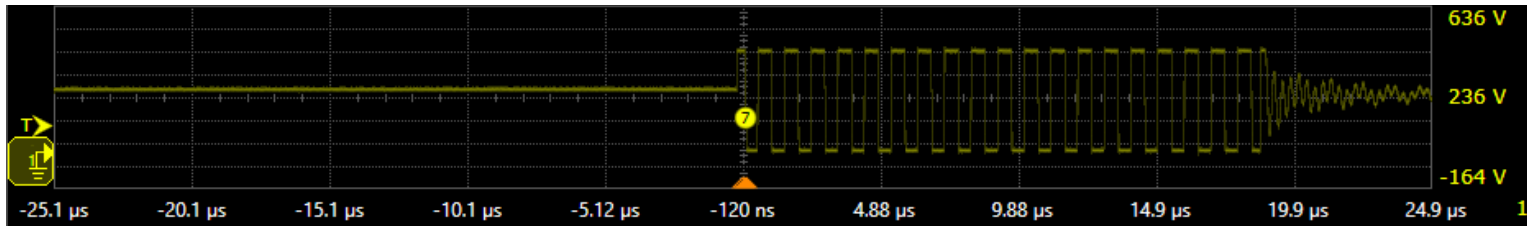
Nominal carrier: 1 MHz

Both branch currents affect switch-node commutation

Burst control regulates the secondary bulk



V1 · 5 V / 3 A



Idle interval

Switching packet

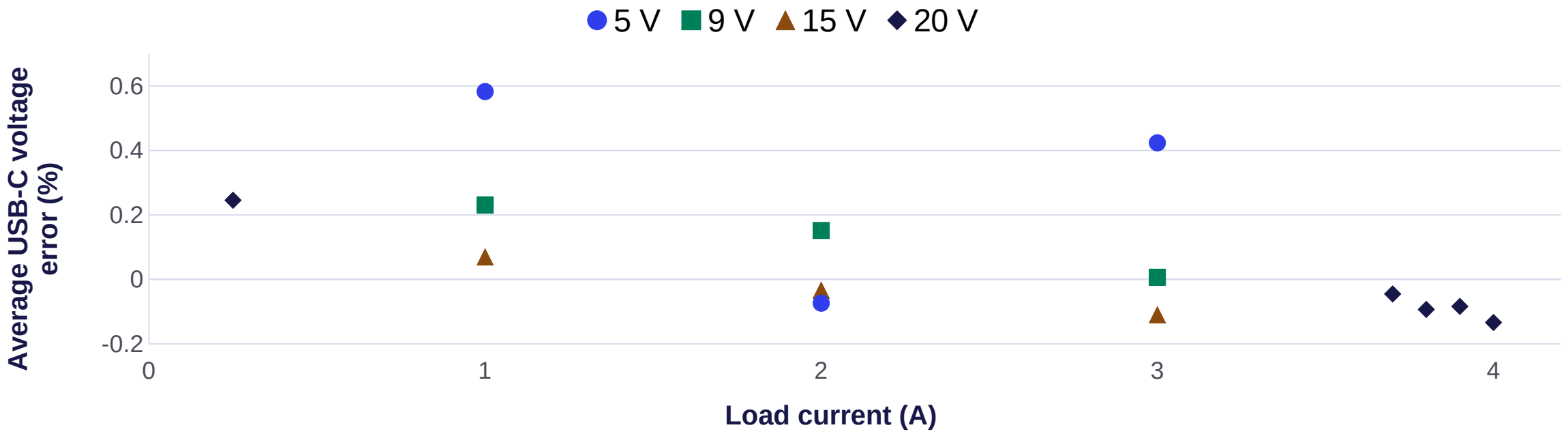
5 µs/div · 100 V/div

Yellow: half-bridge switch-node voltage

- Bursts regulate the secondary bulk voltage
- Energy is transferred in switching packets
- Startup and no-load limits bound the transfer

USB-C voltage follows the selected setting

V1 · 240 V AC, 50 Hz · USB-C connector



Four fixed output settings tested

Largest absolute average error: 0.583% at 5 V, 1 A

79.89 W reached the USB-C connector

V1 · 240 V AC, 50 Hz · 20 V selected

Input: 104.42 W → USB-C connector: 79.89 W

19.9732 V × 4.0 A

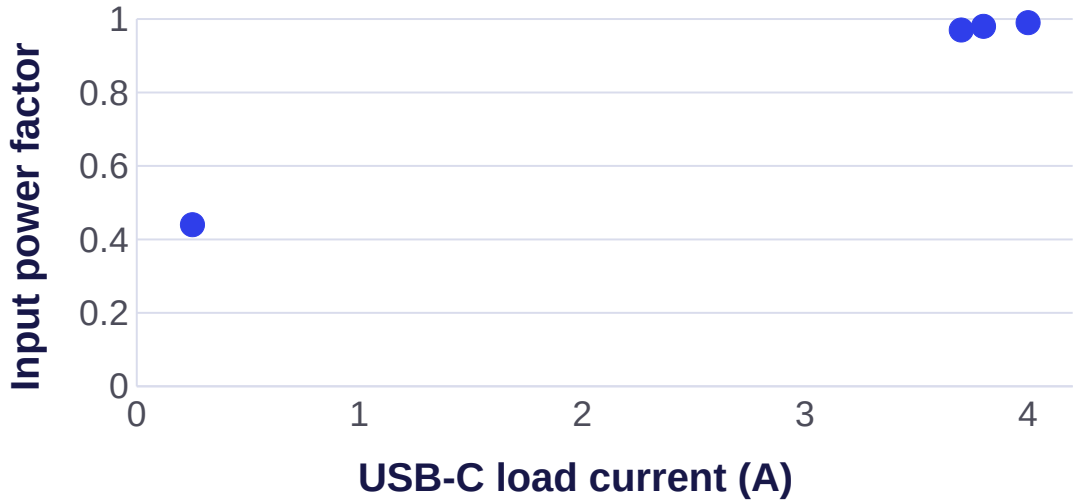
Efficiency estimate: 76.51%

Input–output difference: 24.53 W

Input current improves with load

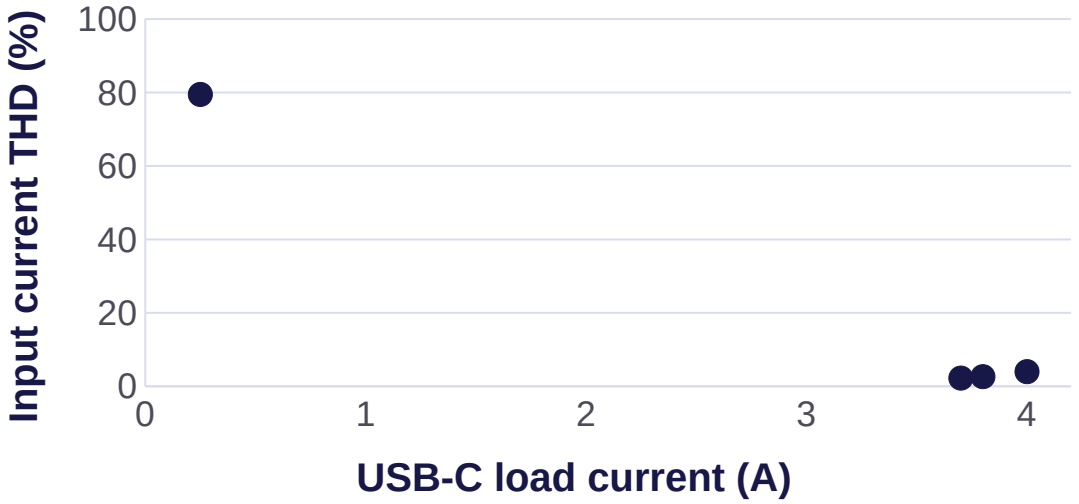
V1 · Selected USB-C output: 20 V

Input power factor



0.44 → 0.99

Input current THD (%)

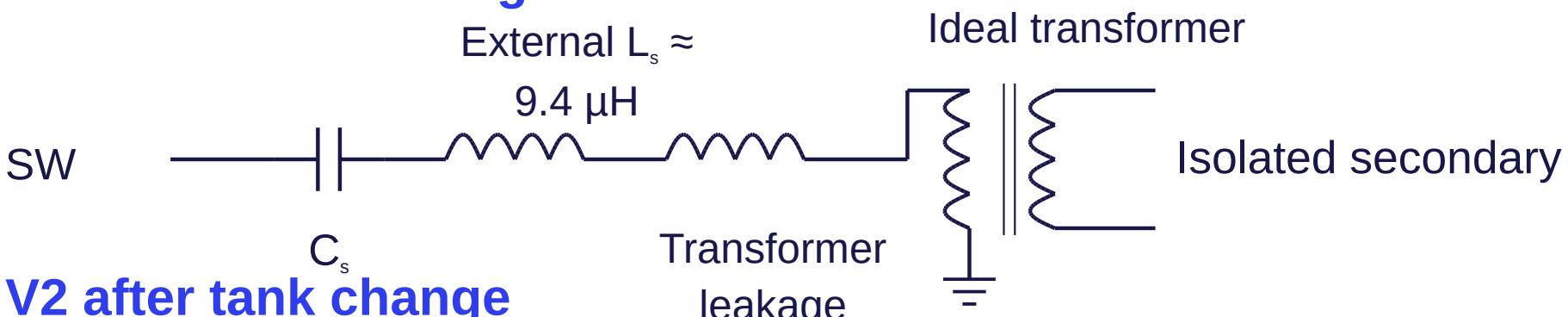


79.46% → 3.98%

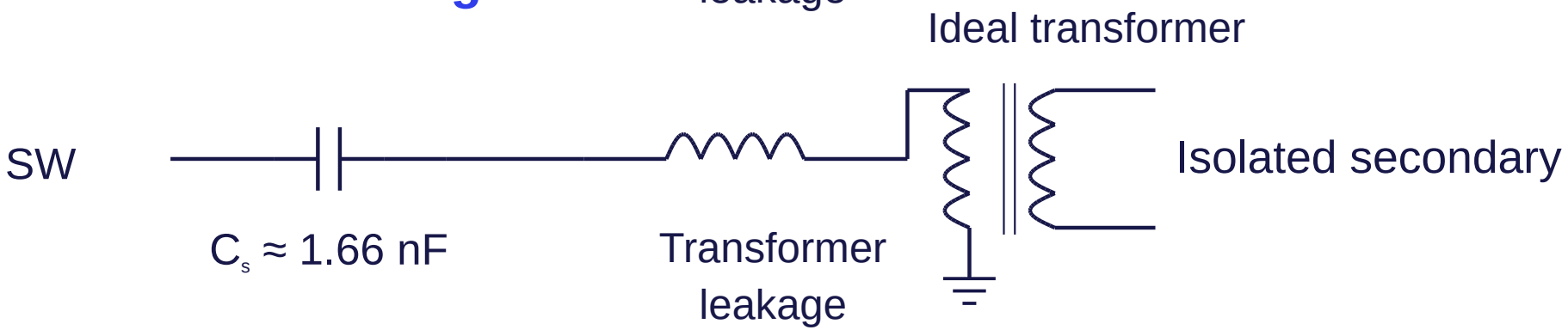
Burst timing can affect input-current shaping

V2 removes the hot series inductor

V2 before tank change



V2 after tank change

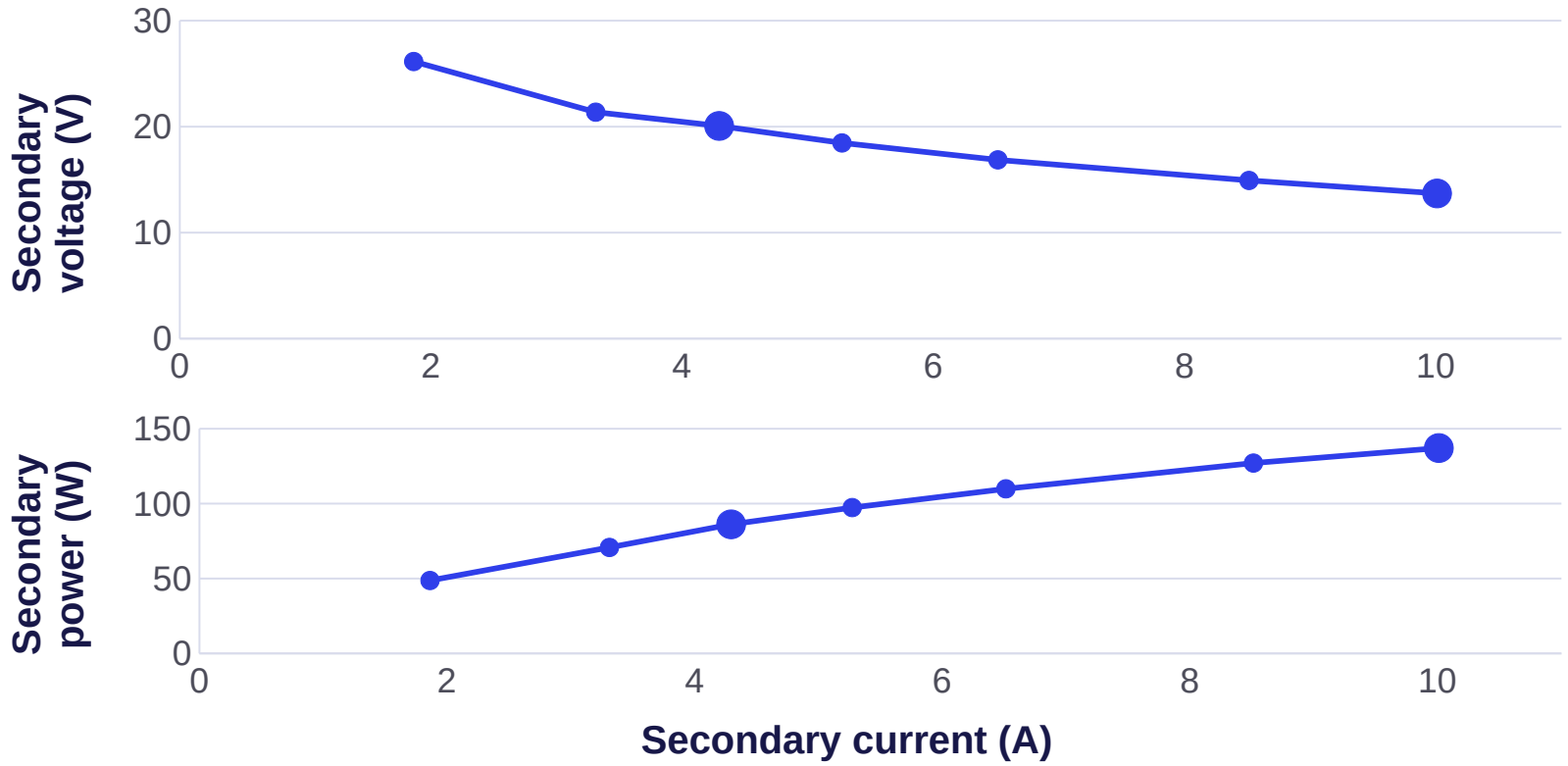


Software dead time ≈ 60 ns

Half-bridge heating fell; input bridge remained hot

More secondary power came with voltage droop

V2 · direct secondary load · 240 V AC, 50 Hz



20.07 V at 4.29 A

13.69 V at 10.01 A

86.17 W at 4.29 A

137.08 W at 10.01 A

Heating remained in the bridge and secondary

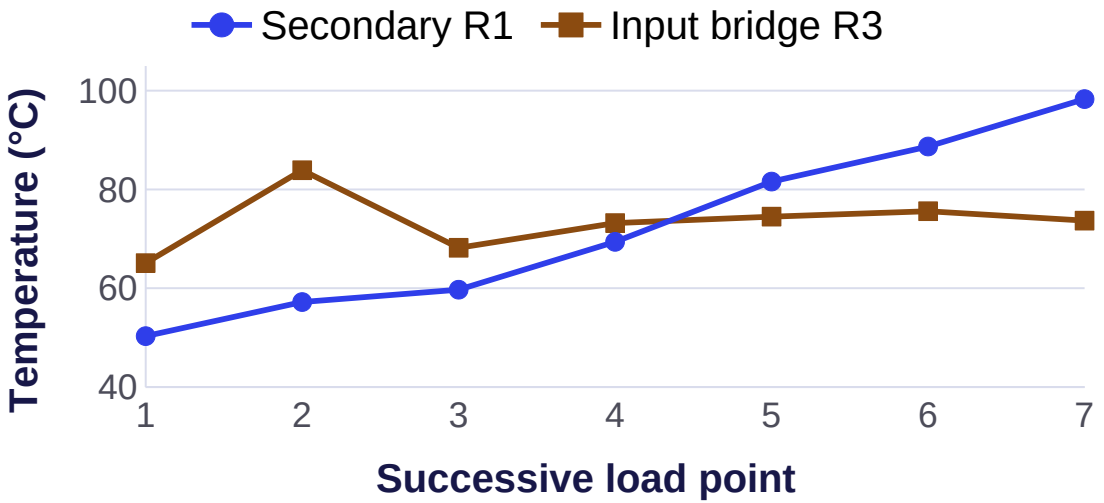
V2 · Electronic load at secondary output

Final point: 137.08 W at secondary



No load point reached thermal equilibrium

Maximum surface temperature by region



Input bridge R3: 83.9 °C at 70.75 W

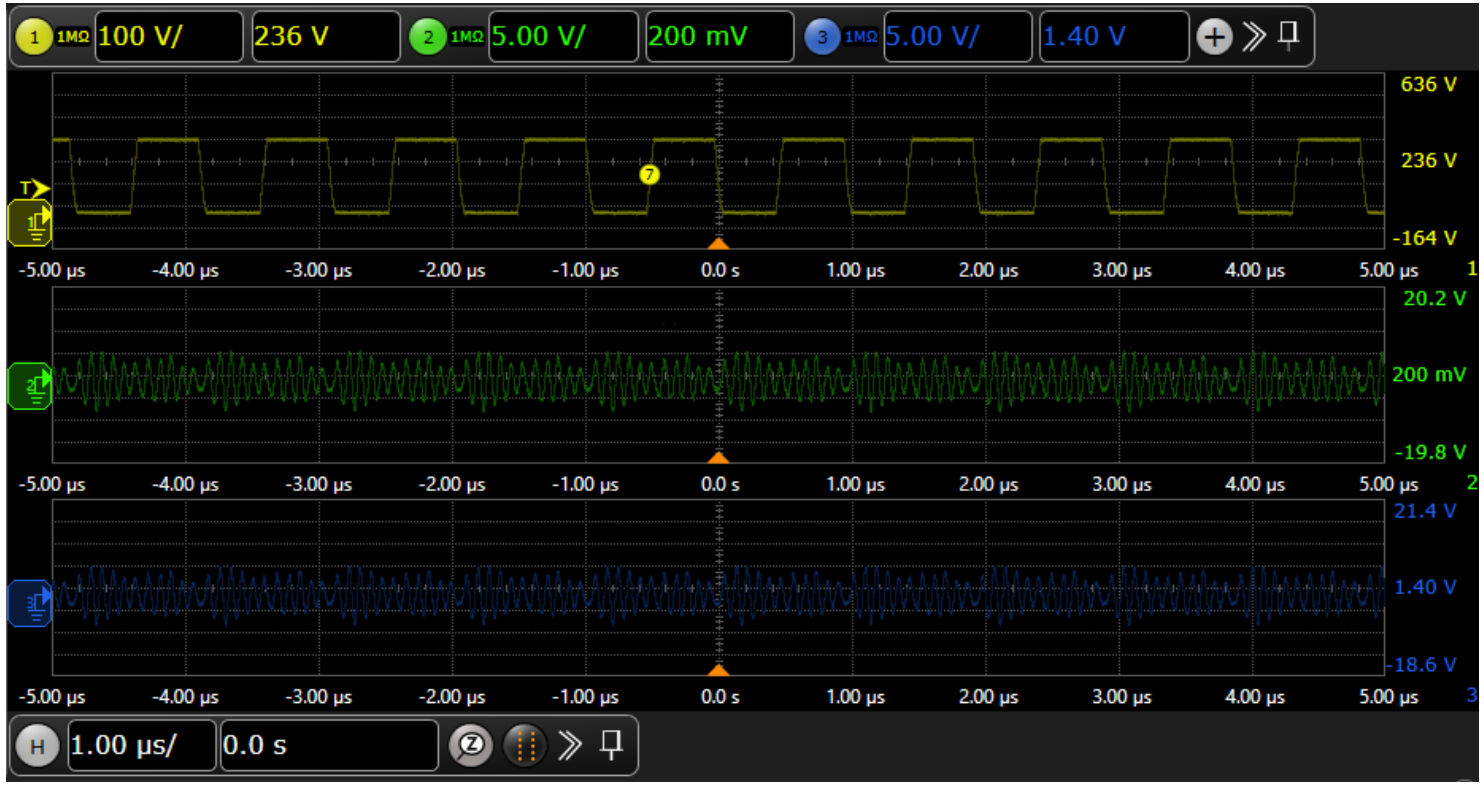
Secondary R1: 98.3 °C at 137.08 W

Continuous USB-C power remains the limit

Implemented	Integrated supply and secondary-voltage burst control
Measured USB-C output	V1: 79.89 W at the connector
Remaining limit	V2: higher secondary power, with voltage droop and heating

V1 switching at the carrier frequency

V1 · 20 V USB-C setting · 4.7 A load



Switching carrier: approximately 1 MHz

Yellow: switch node · DC-coupled · 100 V/div

Green: secondary bulk · AC-coupled · 5 V/div

Blue: USB-C VBUS · AC-coupled · 5 V/div

Time: 1 µs/div

The tank was retuned around transformer leakage

Parameter	Original calculation	Modified V2 calculation
External series inductance	9.4 µH	Removed
Transformer leakage, mean	16.46 µH	16.46 µH
L_{total}: series inductance	25.86 µH	16.46 µH
C_s: series capacitance	1.00 nF	1.66 nF
Ideal series resonance, f_0	0.990 MHz	0.963 MHz

$$f_0 = 1 / (2\pi\sqrt{(L_{total} C_s)})$$

Ideal estimates; installed response also depends on load, magnetizing branch and parasitics.

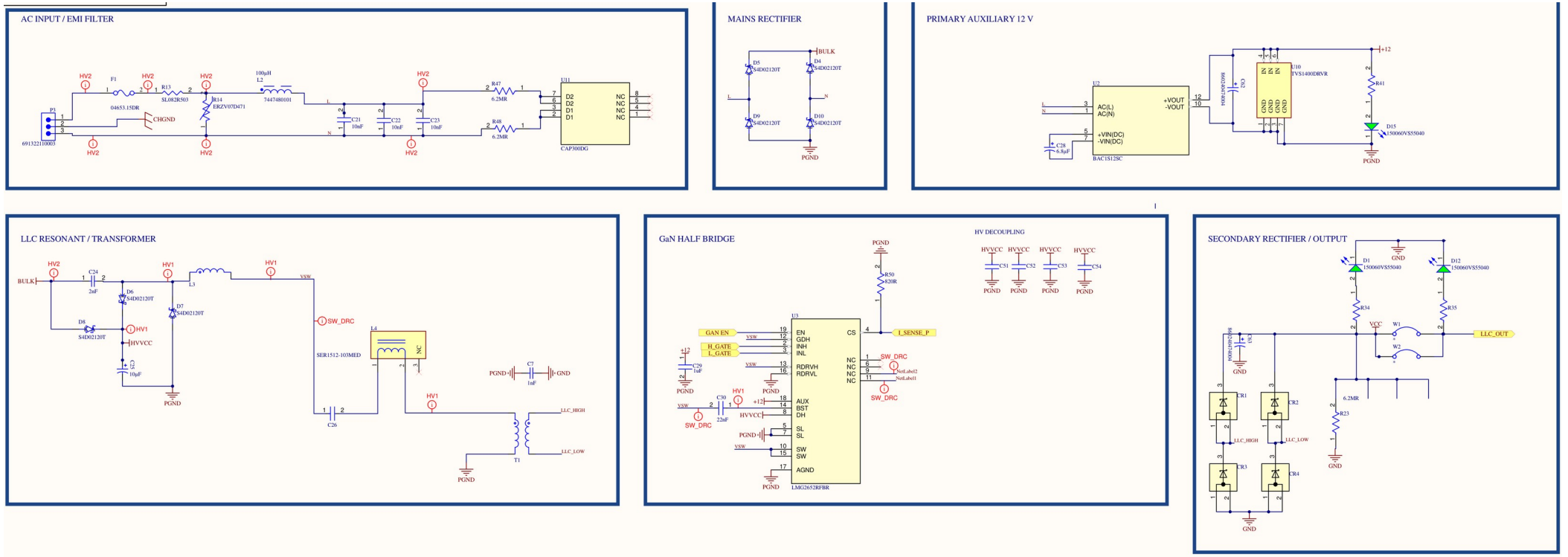
V2 secondary-load measurements

V2 · direct secondary load · 240 V AC, 50 Hz

Point	Secondary voltage (V)	Secondary current (A)	Secondary power (W)	R1 secondary max (°C)	R3 input bridge max (°C)
1	26.14	1.86	48.68	50.3	65.1
2	21.36	3.31	70.75	57.2	83.9
3	20.07	4.29	86.17	59.7	68.2
4	18.46	5.27	97.35	69.4	73.2
5	16.87	6.51	109.87	81.6	74.5
6	14.92	8.51	127.05	88.7	75.6
7	13.69	10.01	137.08	98.3	73.7

Successive load points; no thermal equilibrium

V2 power circuit and earlier work



V2 schematic before the tank change

Ammar et al., IEEE JESTPE, 2020 · doi: 10.1109/JESTPE.2020.2966143

Ammar et al., IEEE JESTPE, 2021 · doi: 10.1109/JESTPE.2021.3083404